17		No with justification	P1	for one correct bound, e.g. 69.5, 70.5, 39.5, 40.5, 121.5, 122.5, 13.5, 14.5
			P1	for complete process to find the upper bound for the volume of the tank, e.g. 120.5 × 40.5 × 70.5 (= 344057.625)
			P1	for complete process to find the upper bound for the number of buckets, (upper bound for volume of tank ÷ lower bound for volume of bucket) e.g. "344057.625" ÷ 13500. Must be in consistent units **OR** correct process to compare the lower bound for 25 buckets of water with the upper bound for the volume of the tank, e.g. 13.5 × 1000 × 25 (= 337500)
			C1	correct conclusion based on correct calculations

Documented scheme clarification

1MA1_MOCK_SET_02_2H_Q17

Original source

scheme page 10, SHA256
73c2d2ed311e18d83d1267b625a2cb420070ca5d64f08801c018ef9003d9c2aa

Original wording

The first bound-example list includes 121.5 and 122.5 for the printed 120 cm length.

Correction / interpretation

The bounds for the 120 cm length are 119.5 cm and 120.5 cm. Use 120.5 x 40.5 x 70.5 = 344057.625 cm3 for the tank upper bound, and 25 x 13500 = 337500 cm3 for the water lower bound. Therefore 25 buckets will not definitely fill the tank. The source first example list misprints the length bounds; its later calculation is correct.

Reason

Rounding to nearest centimetre gives a half-centimetre error on either side of 120.

SoI authority

51c4d6e17933058a9fe81587949167051dbebe1efcfc61bb70dea124233e14c3